

- 14 An ideal monatomic gas has 1000 J of heat added to it and it does 500 J of work; its thermodynamic temperature changes by ΔT_1 . When twice the amount of heat is added to it and it does the same amount of work, its temperature changes by ΔT_2 . The ratio of $\Delta T_1 / \Delta T_2$ is

A 1/5

B 1/3

C 3/5

D 1

Ans : B

$$\begin{aligned}\Delta U_1 &= Q_1 + W_1 \\ &= 1000 - 500 \\ &= 500 \text{ J}\end{aligned}$$

$$\begin{aligned}\Delta U_2 &= Q_2 + W_2 \\ &= 2000 - 500 \\ &= 1500 \text{ J}\end{aligned}$$

$$\Delta U \propto \Delta T$$

$$\frac{\Delta U_1}{\Delta U_2} = \frac{T_1}{T_2}$$

$$\frac{T_1}{T_2} = \frac{500}{1500} = \frac{1}{3}$$

15 One end of a spring is fixed to a support. A block of mass 2.0 kg is attached to the other end of